Weekly Progress Report

Mathematical Study of Radar Scattering from Heterogeneous Rough Soil Using the Finite Element Method

Week 1

Abstract

This week's work focused on developing the mathematical foundations required to understand the finite-element modelling of electromagnetic scattering from rough and heterogeneous soil. The study is based on the paper "Computation of Radar Scattering From Heterogeneous Rough Soil Using the Finite-Element Method". The main objective at this stage was not to reproduce the numerical experiments, but to understand the underlying mathematical problem and how the finite-element method (FEM) can be used to convert the continuous electromagnetic boundary-value problem into a finite-dimensional system of equations. The report first introduces radar scattering, soil moisture, surface roughness and dielectric properties, and then develops the FEM methodology from the basic idea of discretization through the weak formulation, Galerkin method, basis-function representation, element matrices, global assembly, boundary conditions and the final linear system.

1 Introduction and Objective

The problem being studied is the computation of radar scattering from a rough soil surface whose electromagnetic properties may vary with depth. The problem is relevant to radar-based sensing of soil moisture because the electromagnetic response of soil depends strongly on its dielectric properties, which are themselves related to moisture content.

The paper considers a two-dimensional representation of the physical problem in which the soil surface is described by

$$z = f(x).$$

Thus, the surface is rough in one spatial direction while the electromagnetic field is solved in a two-dimensional computational domain. This provides a model that is substantially less expensive than a fully three-dimensional simulation while still allowing surface roughness to be incorporated.

The main mathematical chain underlying the problem can be summarized as

$$\boxed{m_v \longrightarrow \epsilon_r \longrightarrow \text{Maxwell's equations} \longrightarrow \text{wave equation} \longrightarrow \text{FEM discretization} \longrightarrow \text{electromagnetic fi}}$$

The purpose of this week's work was to understand the first several stages of this chain, especially how the electromagnetic partial differential equation is transformed into a finite-dimensional algebraic problem using FEM.

2 1. Physical Problem

2.1 Radar scattering

Radar operates by transmitting an electromagnetic wave toward an object or surface and measuring the resulting scattered electromagnetic field. When the incident wave reaches the soil surface, the field interacts with the material and part of the incident energy is reflected, transmitted into the soil, and scattered into different directions.

The part of the scattered radiation that travels back toward the radar is referred to as *backscatter*. Thus, the basic physical process can be represented as

$$\text{incident wave} \rightarrow \text{interaction with soil} \rightarrow \text{scattered electromagnetic field.}$$

The quantity ultimately of interest is the radar scattering strength, which is described using the radar cross section (RCS).

2.2 Soil moisture

Let m_v denote volumetric soil moisture. Conceptually,

$$m_v = \frac{\text{volume of water}}{\text{volume of soil}}.$$

The presence of water significantly changes the electromagnetic properties of soil. In particular, the relative permittivity ϵ_r depends on soil moisture.

The paper uses an empirical dielectric model of the form

$$\epsilon_r = \epsilon_r(m_v).$$

For the silt-loam model considered in the paper,

$$\epsilon_r(m_v) = \left(2.201 + 26.406m_v + 34.563m_v^2\right) - \left(-0.141 + 11.820m_v + 8.100m_v^2\right) j.$$

The important mathematical relationship is therefore

$$\boxed{m_v \longrightarrow \epsilon_r.}$$

If the moisture content varies with depth, then

$$m_v = m_v(z),$$

and hence

$$\epsilon_r = \epsilon_r(z).$$

Therefore, the electromagnetic coefficients in the governing differential equation may themselves be functions of position.

2.3 Homogeneous and heterogeneous soil

For homogeneous soil,

$$m_v(z) = m_0,$$

and therefore

$$\epsilon_r(z) = \epsilon_{r,0}.$$

The material properties are constant throughout the computational domain.

For heterogeneous soil,

$$m_v = m_v(z),$$

so that

$$\epsilon_r = \epsilon_r(z).$$

A conceptual depth-dependent profile can therefore be represented as

$$m_v(z) = \begin{cases} \text{varying function of } z, & 0 < z < d, \\ m_0, & z \geq d, \end{cases}$$

where d is the depth over which the moisture profile changes.

This makes the problem more complicated because the coefficients of the electromagnetic PDE are no longer constant.

3 2. Geometrical Representation of Rough Soil

A perfectly flat soil-air interface would be represented by

$$z = 0.$$

The paper instead considers a rough interface represented by

$$\boxed{z = f(x)}.$$

This means that the height of the soil surface changes as a function of the horizontal coordinate x.

A random rough surface can be statistically characterized using parameters such as the root-mean-square height h and the correlation length l. For a Gaussian-correlated surface, the RMS slope is

$$s = \frac{\sqrt{2}h}{l}.$$

Here,

- h measures the typical magnitude of surface-height fluctuations,
- l measures the characteristic horizontal scale over which the surface varies,
- s provides a measure of the steepness of the roughness.

The paper generates random surfaces with specified statistical properties and uses Monte Carlo realizations when studying ensemble behaviour.

4 3. Electromagnetic Formulation

4.1 Maxwell's equations

The electromagnetic fields are described by Maxwell's equations. Under a time-harmonic convention, the source-free equations may be written as

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H},$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E},$$

together with the corresponding divergence conditions

$$\nabla \cdot (\epsilon \mathbf{E}) = 0, \qquad \nabla \cdot (\mu \mathbf{H}) = 0$$

in a source-free region.
Here,

- $\mathbf{E}$ is the electric field,
- $\mathbf{H}$ is the magnetic field,
- ϵ is the permittivity,
- μ is the permeability,
- ω is the angular frequency.

The objective is to determine the electromagnetic field inside the computational domain when an incident radar wave is applied.

4.2 Wave equation

By eliminating one of the electromagnetic fields from Maxwell's equations, a vector wave equation can be obtained. In the formulation used by the paper, the relevant field is denoted by U.

The general structure of the wave equation can be written as

$$\nabla \times \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q U = 0,$$

where k_0 is the free-space wavenumber and the coefficients p and q depend on the polarization.

For TM polarization,

$$U = H_y, \qquad p = \epsilon_r, \qquad q = \mu_r,$$

while for TE polarization,

$$U = E_y, \qquad p = \mu_r, \qquad q = \epsilon_r.$$

Therefore, the soil moisture enters the wave equation indirectly through the position-dependent material parameter

$$\epsilon_r = \epsilon_r(m_v(z)).$$

5 4. Why a Numerical Method is Necessary

For simple geometries and constant material properties, analytical solutions may sometimes be obtained. However, the present problem has several sources of mathematical complexity:

1. the surface geometry is rough and may be represented by a complicated function $z = f(x)$;
2. the material coefficient ϵ_r may depend on position;
3. the electromagnetic wave equation is a partial differential equation;
4. the physical problem is effectively unbounded, requiring appropriate boundary conditions;
5. the field must satisfy electromagnetic interface conditions at the rough soil-air boundary.

Because of these factors, obtaining a closed-form analytical solution is generally difficult. A numerical method is therefore required.

The paper uses the finite-element method to obtain an approximate solution.

6 5. Finite Element Method: Fundamental Idea

The finite-element method is a numerical method for solving differential equations by replacing a continuous problem with a finite-dimensional approximation.

The fundamental idea is to divide the computational domain into a finite number of small subdomains called *elements*.

For a two-dimensional problem, triangular elements are particularly useful because they can represent complicated geometries and rough boundaries.

Thus,

$$\boxed{\text{complicated continuous domain} \longrightarrow \text{many small elements.}}$$

The points connecting these elements are called *nodes*.

For example, a two-dimensional domain can be represented schematically as

$$\Omega = \bigcup_{e=1}^{N_e} \Omega_e,$$

where Ω_e denotes the e-th element.

The FEM approximation does not attempt to find the exact field at every point directly. Instead, it approximates the field using a finite number of unknown coefficients associated with the basis functions.

6.1 Finite-dimensional approximation

Let $U(\mathbf{r})$ denote the exact electromagnetic field.

The FEM approximation is written as

$$U_h(\mathbf{r}) = \sum_{i=1}^{N} u_i N_i(\mathbf{r}),$$

where

- $N_i(\mathbf{r})$ are basis functions,
- u_i are unknown coefficients,
- N is the total number of degrees of freedom.

Thus, instead of solving for an arbitrary continuous function $U(\mathbf{r})$, we only need to determine the finite vector

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \cdots & u_N \end{bmatrix}^T .$$

This is the fundamental reduction performed by FEM:

$$\boxed{\text{infinite-dimensional PDE problem} \longrightarrow \text{finite-dimensional linear algebra problem.}}$$

7 6. Basis Functions and Local Approximation

The field is approximated locally on each finite element.

For a simple one-dimensional linear element with endpoints x_1 and x_2, the approximate solution can be written as

$$u(x) = N_1(x)u_1 + N_2(x)u_2,$$

where

$$N_1(x) = \frac{x_2 - x}{x_2 - x_1},$$

and

$$N_2(x) = \frac{x - x_1}{x_2 - x_1}.$$

These functions satisfy

$$N_1(x_1) = 1, \qquad N_1(x_2) = 0,$$

and

$$N_2(x_1) = 0, \qquad N_2(x_2) = 1.$$

Thus, the value of the approximate solution at a node is exactly the corresponding nodal unknown.

The same principle is extended to two-dimensional elements. In the electromagnetic formulation of the paper, first-order Whitney elements are used as basis functions.

The exact form of the Whitney basis will be studied in a later stage; at the present stage, the important point is that they provide suitable local basis functions for representing electromagnetic fields on a triangular mesh.

8 7. Weak Formulation

The next major step is to convert the differential equation into a weak or variational formulation.

Suppose, abstractly, that the governing PDE is

$$L(U) = 0.$$

If an approximate FEM field U_h is substituted into the differential equation, it will generally not satisfy the PDE exactly at every point. Define the residual

$$R = L(U_h).$$

Instead of requiring

$$R(\mathbf{r}) = 0$$

pointwise, FEM requires a weighted residual to vanish.

For a test function T,

$$\int_\Omega T\, R\, d\Omega = 0.$$

This is the basic idea behind a weak formulation.

The advantage is that the differential equation can be transformed into an integral equation that can be naturally applied over individual finite elements.

8.1 Weak form of the electromagnetic problem

For the electromagnetic problem studied in the paper, the weak formulation has the structure

$$\int_\Omega \left[(\nabla \times T) \cdot \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q\, T \cdot U \right] dS = \text{boundary terms.}$$

The appearance of the boundary term is a consequence of applying vector identities and Green's theorem to the differential equation.

This weak formulation is the starting point for the finite-element discretization.

9 8. Galerkin Method

A weak formulation alone does not yet give a finite set of numerical equations. We now substitute the finite-dimensional approximation

$$U_h(\mathbf{r}) = \sum_{i=1}^{N} u_i N_i(\mathbf{r}).$$

The Galerkin method chooses the test functions from the same finite element space. Thus, we may choose

$$T = N_j$$

for $j = 1, 2, \ldots, N$, or the corresponding vector-valued basis functions for the electromagnetic problem.

Substitution gives

$$\int_\Omega \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times \sum_{i=1}^{N} u_i N_i \right) - k_0^2 q N_j \cdot \sum_{i=1}^{N} u_i N_i \right] dS = \text{boundary contribution.}$$

Using linearity,

$$\sum_{i=1}^{N} u_i \int_\Omega \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS = \text{boundary contribution.}$$

This produces one equation for each j.

Therefore, the system can be written in matrix form as

$$\boxed{A\mathbf{u} = \mathbf{b}.}$$

This is the central mathematical result of the FEM discretization.

10 9. Element Matrices

The integral expressions above can be evaluated separately over each finite element.

Let Ω_e denote the e-th triangular element.

The contribution from one element can be represented by an element matrix $A^{(e)}$, whose entries have the general form

$$A_{ij}^{(e)} = \int_{\Omega_e} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS + B_{ij}^{(e)},$$

where $B_{ij}^{(e)}$ represents any contribution arising from boundary terms associated with the element.

The important feature is that the integral is computed only over one small element.

Thus, instead of directly constructing an enormous global matrix, FEM first computes many small local matrices.

Schematically,

$$\boxed{\text{PDE} \to \text{weak form} \to \text{element equations} \to \text{element matrices.}}$$

11 10. Global Assembly

The elements are not independent. Neighbouring elements share nodes, and therefore their local contributions must be combined.

Suppose two elements share a node. The unknown associated with that node is common to both elements. Therefore, the equations from both elements contribute to the same global degree of freedom.

The local element matrices are assembled into one global matrix:

$$A = \sum_{e=1}^{N_e} A^{(e)}$$

in the appropriate global indexing sense.

The resulting global system is

$$A\mathbf{u} = \mathbf{b}.$$

This is known as the *assembly process.*

The important conceptual distinction is

$$\boxed{\text{local element matrices} \longrightarrow \text{one globally coupled system.}}$$

Although each element represents only local interactions, the global system contains the coupling between the entire computational domain.

12 11. Sparsity of the Global Matrix

A major computational advantage of FEM is that the global matrix is sparse.

This occurs because the basis functions are local. A basis function is nonzero only over a small number of neighbouring elements.

Consequently, two degrees of freedom corresponding to distant regions of the domain generally do not interact directly.

Thus, most matrix entries satisfy

$$A_{ij} = 0.$$

The global matrix therefore has the schematic structure

$$A = \begin{bmatrix} * & * & 0 & 0 & \cdots \\ * & * & * & 0 & \cdots \\ 0 & * & * & * & \cdots \\ 0 & 0 & * & * & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

The paper takes advantage of this property and solves the resulting sparse system using a sparse direct solver.

For large electromagnetic meshes, the number of unknowns can become very large, so exploiting sparsity is essential for computational feasibility.

13 12. Mesh Resolution

The finite-element mesh must be sufficiently fine to resolve the electromagnetic wavelength.

The paper uses a mesh-size restriction of approximately

$$l_{\max} < \frac{\lambda}{20},$$

where λ is the wavelength in the medium.

This requirement has a clear interpretation. If an element is much larger than the wavelength, the electromagnetic field may vary significantly within one element, making a low-order approximation inaccurate.

Therefore,

$$\text{smaller elements} \rightarrow \text{better spatial resolution},$$

but also

$$\text{smaller elements} \rightarrow \text{more degrees of freedom} \rightarrow \text{higher computational cost}.$$

Thus, the mesh represents a balance between numerical accuracy and computational efficiency.

14 13. Boundary Conditions in the FEM Model

The physical radar-scattering problem is effectively unbounded. However, a computer can only represent a finite computational domain.

Consider a simplified domain

$$\Omega = \Omega_{\text{air}} \cup \Omega_{\text{soil}}.$$

The outer boundary of the computational domain is artificial. If a normal reflecting boundary were used, scattered waves would reflect from this artificial boundary and contaminate the physical solution.

To avoid this, the paper uses first-order absorbing boundary conditions (ABCs).

The purpose of the ABC is to approximate an outgoing-wave condition:

$$\boxed{\text{outgoing scattered wave} \rightarrow \text{leave computational domain}}$$

without significant reflection back into the domain.

The paper formulates the boundary condition in terms of the scattered field in the air region.

It is important to understand that the boundary condition enters the weak formulation and therefore modifies the final matrix system.

Schematically,

$$A = A_{\text{volume}} + A_{\text{boundary}}.$$

The paper also discusses the fact that first-order ABCs are not perfectly reflectionless, so they constitute one source of numerical error.

15 14. Incident and Scattered Fields

The total electromagnetic field may be written conceptually as

$$U = U_i + U_s,$$

where

- U_i is the incident field,
- U_s is the scattered field.

In the air region, the paper separates the known incident field from the total field and applies the absorbing boundary condition to the scattered field.

This is useful because the objective is to determine the field produced by the interaction of the incident radar wave with the soil.

Thus, the numerical calculation can be viewed as

$$\boxed{\text{known incident field} + \text{unknown scattered response.}}$$

The paper uses a tapered incident field so that the illumination is reduced near the lateral boundaries of the surface. This reduces edge effects associated with a finite computational domain.

16 15. Complete FEM Procedure for the Radar Problem

The complete procedure can now be summarized mathematically.

Step 1: Specify the physical parameters

Define

$$\lambda, \qquad \theta_i, \qquad h, \qquad l, \qquad m_v(z),$$

together with the corresponding material model.

Step 2: Define the rough geometry

Construct the soil-air interface

$$z = f(x).$$

The resulting computational domain is

$$\Omega = \Omega_{\text{air}} \cup \Omega_{\text{soil}}.$$

Step 3: Define material properties

Using the moisture model,

$$\epsilon_r(z) = \epsilon_r(m_v(z)).$$

The values of ϵ_r are therefore assigned throughout the soil domain.

Step 4: Generate the mesh

Divide Ω into triangular elements:

$$\Omega = \bigcup_{e=1}^{N_e} \Omega_e.$$

Choose the element size sufficiently small compared with the wavelength.

Step 5: Choose basis functions

Represent the electromagnetic field as

$$U_h(\mathbf{r}) = \sum_i u_i N_i(\mathbf{r}).$$

For the electromagnetic problem, suitable vector basis functions such as first-order Whitney elements are employed.

Step 6: Write the weak formulation

For every test function T, impose

$$\int_\Omega \left[(\nabla \times T) \cdot \left(\frac{1}{p} \nabla \times U_h \right) - k_0^2 q\, T \cdot U_h \right] dS = \text{boundary contributions}.$$

Step 7: Apply Galerkin discretization

Choose test functions from the same finite-element space and substitute

$$U_h = \sum_i u_i N_i.$$

This gives algebraic equations for the unknown coefficients u_i.

Step 8: Compute local element matrices

For each element Ω_e, evaluate the required integrals to obtain

$$A^{(e)}$$

and the corresponding local right-hand-side contribution.

Step 9: Assemble the global system

Combine all local contributions:

$$A\mathbf{u} = \mathbf{b}.$$

Step 10: Impose boundary conditions

Include the contributions from the absorbing boundary conditions and the incident excitation.

Step 11: Solve the sparse linear system

The unknown vector

$$\mathbf{u}$$

is obtained by solving

$$A\mathbf{u} = \mathbf{b}.$$

This gives the coefficients of the finite-element approximation to the electromagnetic field.

Step 12: Reconstruct the electromagnetic field

Once the coefficients u_i are known,

$$U_h(\mathbf{r}) = \sum_i u_i N_i(\mathbf{r})$$

provides the approximate electromagnetic field throughout the computational domain.

Step 13: Calculate the scattered field and RCS

The scattered field is extracted from the FEM solution and transformed to a far-field quantity. The radar cross section is then computed as a measure of the scattering strength.

Thus, the overall computational pipeline is

$$\boxed{\begin{aligned}&m_v(z) \to \epsilon_r(z) \to \text{rough geometry} \to \text{mesh}\\ &\to \text{basis functions} \to \text{weak form} \to \text{element matrices}\\ &\to \text{global matrix} \to A\mathbf{u} = \mathbf{b} \to U_h \to U_s^{\text{far}} \to \sigma.\end{aligned}}$$

17 16. Why FEM is Appropriate for this Problem

The finite-element method is particularly appropriate for the problem because the geometry and material coefficients can be complicated.

First, the rough boundary

$$z = f(x)$$

can be represented naturally by an unstructured triangular mesh.

Second, the dielectric constant can vary spatially:

$$\epsilon_r = \epsilon_r(x, z).$$

Such variation can be incorporated directly into the element integrals.

Third, the method does not require the entire physical domain to have a simple analytical geometry.

Finally, the local structure of finite elements leads to a sparse global matrix, making large numerical calculations possible.

Thus, FEM converts the complicated electromagnetic problem into a systematic sequence of local calculations followed by a global linear solve.

18 17. Validation Strategy

The paper does not rely only on the FEM solution. Numerical methods should be checked against independent methods whenever possible.

For relatively smooth surfaces, the first-order Small Perturbation Method (SPM) provides an independent reference. The first-order SPM is applicable in a restricted regime such as

$$kh < 0.3, \qquad s < 17^\circ.$$

Therefore, FEM can first be compared with SPM in a parameter regime in which the latter is expected to be accurate.

For flat soil with depth-dependent moisture, the paper also compares FEM with the transfer matrix method, where the heterogeneous soil is approximated by many thin layers.

This suggests a useful general validation strategy:

$$\boxed{\text{FEM} \quad \text{vs.} \quad \text{independent mathematical model}}$$

before using FEM for more complicated cases.

19 18. Current Progress and Next Steps

The work completed during this initial stage has focused on building a mathematical understanding of the problem rather than reproducing the full numerical study.

The main concepts established are:

1. formulation of the soil surface as

$$z = f(x);$$

2. understanding of soil moisture as a quantity influencing the relative permittivity

$$\epsilon_r = \epsilon_r(m_v);$$

3. recognition that depth-dependent moisture produces

$$\epsilon_r = \epsilon_r(z);$$

4. formulation of the electromagnetic problem through Maxwell's equations and the associated wave equation;

5. understanding of the weak formulation of the electromagnetic PDE;
6. understanding of the basic FEM approximation

$$U_h = \sum_i u_i N_i;$$

7. understanding of the Galerkin procedure;
8. derivation of the conceptual matrix form

$$A\mathbf{u} = \mathbf{b};$$

9. understanding of triangular meshes, local element matrices, global assembly and sparse matrices;
10. understanding of the role of absorbing boundary conditions.

The next stage will focus on carrying out the mathematical derivation in greater detail, particularly the derivation from Maxwell's equations to the weak formulation and the explicit construction of the finite-element matrices for a triangular electromagnetic element.

The subsequent stages will then address numerical implementation, far-field extraction, radar cross section calculation, and comparison with the reference methods used in the paper.

20 Conclusion

The central understanding developed during this week is that FEM is not a separate physical model for radar scattering. Rather, it is a numerical framework for solving the governing electromagnetic boundary-value problem.

The mathematical structure is

$$\boxed{\text{physical model} \to \text{PDE} \to \text{weak formulation} \to \text{finite-element approximation} \to \text{linear algebra.}}$$

For the present application, this becomes

$$\boxed{\text{soil moisture} \to \epsilon_r \to \text{Maxwell} \to \text{electromagnetic wave equation} \to \text{FEM} \to A\mathbf{u} = \mathbf{b}.}$$

Understanding this chain provides the mathematical foundation required for the next stage of the project, in which the finite-element formulation will be derived explicitly and eventually implemented for the rough-soil radar-scattering problem.